

ActInf Textbook Group ~ Cohort 5 ~ Onboarding & Overview (September 2023)

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Hello, it's September 17th, 2020-23. This video is an overview and an onboarding for the textbook group, specifically Cohort 5 of the Active Inference textbook group at the Active Inference Institute. The book we're discussing is Active Inference, the Free Energy Principle in Mind, Brain, and Behavior by Thomas Parr, Giovanni Pizzullo, and Carl Friston. You can get a hardcover book, or as we'll show in a second, this is an open source book so we can share the PDF for free.

This video is mainly going to be an overview of the sections of this coda, which represent the asynchronous participation, or at least a part of it, for the textbook group. Also, you can do whatever work you want in your own workspaces, but as we'll see, there are some cool features of having a shared epistemic niche or workspace that enable us to add a lot of value and connect a lot of dots for each other, and we'll talk about synchronous meetings as well. However, interacting with this document is where all the information is kept. A directory page covers a few different areas and pages that you might want to look at. In the cohorts and meeting section, also this is a total living document just showing a snapshot at one time. Cohort 5 is just beginning in the coming week. And just to give a little bit of a scatter plot about people's background, in this figure, on the x-axis is people's self-reported familiarity with active inference across different registration periods of the textbook group. And on the y-axis are people's self-evaluated familiarity with green, computer science, blue, biology, and red, math. And so it just gives a little bit of an idea so everyone, all learning styles and approaches that people are going to take will be welcomed and hopefully people can participate however is meaningful for them. In the page Cohort 5, Live Meetings, there's a table that has the dates and the times of the meetings that we'll have that'll be recorded for Cohort 5. After the meetings, the video recording will be listed in this column as a clickable link. For example, in Cohort 4, this is what that looks like. In the first week, we're going to have an onboarding and an overview. Hopefully some people will have watched this video by then, however, it will be okay if not. Following that, we'll have two weeks per chapter going through the first five chapters. And the meetings alternate times, so people may or may not be able to join one or both sessions synchronously. Cohort 4, who will be going through the second half of the textbook, have alternating times in the other direction. So if you prefer to just visit one time at a weekly rate and hear about different parts of the textbook since we're referencing and discussing and building on shared material, then you're welcome to come to really either of the ongoing series of synchronous meetings, or watch the video recordings. And we'll go through a few more of these top sections and then discuss how different aspects of the book can be referenced in our questions and in our learning. In the onboarding section, there's a little bit more information about participation modes and some sections on learning practices. I won't go into them in this video, but people have added many different ways and we can always add more ways that people felt like they were able to be actively learning. Different ways of learning. In this section, Parr, Pizzullo, and Friston 2022 textbook, several sections of the textbook or features of it

have been already brought into this CODA document. So in the chapter section, there's a list of the chapters and a detailed table of contents. Unfolding the chapters section takes you to a page for each chapter. And for most of the chapters or all the chapters, we have an overview video, maybe a 15 to 25 minute video of the chapter. So we have myself and Ali giving a bit of an overview of the text of the chapter itself. Then there's a table of questions. And for each question, there are ways to upvote it as interesting and rise those up in the discussion. And also there's an expandable section where discourse can exist on each question and reference a lot of other things. So questions are a big part of the textbook group. It's the minimal epistemic action. And then there's a lot of discussion that could be brought into that questions table. Otherwise, just sort of open-ended discussion and notes and questions that people have brought in. So for each chapter, we center around this and in the synchronous discussions, look at the questions that people bring up. The equation section, you can use control plus and minus or control and mouse scroll to zoom in or out in Coda. In the equation section, there's a table with all of the equations with screenshots and latex rendering and plain text and also descriptions in discourse. For example, equation 2.5 with variational free energy. Also, the active inference ontology is used to give a plain text reading of the equations. And then there's different discussions about aspects of the equations. Of course, everyone's welcome to add their own pieces. Not all of them are as built out as some.

Also, there's some experiments into using the language models in Coda and the latex plain text to give math explanations, explaining it to high school student, writing code, explaining where it might be important, explaining the variables, asking philosophy questions, and also translating it into different languages.

And we can add more languages. The figures page has a table with all the figures. This is very commonly we want to reference a specific figure. And Coda allows us to do that by using at and then typing figure and then figure 4.3 becomes something that we can bring up and that becomes really useful.

So, in the example, the boxes and table. Boxes and table similarly covers the boxes and table. And code has some discussion and also several dozen implementations of code for active inference models for people who want

to explore that. So that's largely bringing features about the textbook in. In the discourse section, there's a lot of additional building and resources that successive cohorts continue to work on. So there's an underlying large questions table that's also broken out in views by cohort. So if you want to make sure that a questions addressed in a cohort 5 meeting, then adding it to this table, it'll all feed into the same underlying view,

but that'll help us prioritize those discussions. And if you want to add a new row, you can click into an existing row and just, then just add enter. And everyone is welcome to add questions, ideally before the discussion, so that we can

review each other's questions and upvote ones that are interesting. Also begin to add our own thoughts and relevant resources and further questions in the discussion section as we're largely building on each other's collaboration asynchronously. The weekly meetings are, for several months, one to two hours per week.

However, it's really the work that people put in, in terms of the questions that they ask and the practices

that they engage in outside of the synchronous meetings. And so asking questions and building on others and sharing our perspectives that way is one of the most important ways to do that. Depending on interest, there's a lot of work in math learning groups on various features that people can explore. And this has various ways that it's been approached throughout the cohorts, ranging from background and fundamental aspects of math, all the way on through a lot of rigor and connecting the dots between equations in the textbook. In projects, people have listed projects that they want to or are or would work on. And a lot of them have come through. A lot of people have also done things that aren't shown here, but it's a place to get visibility for your project if you're looking for collaboration or anything like that. Ideas and insights also can be added to row by row. They don't need to be questions, just any thoughts that you have. Resources has some different resources that supplement or complement the textbook itself. Future textbook group

has an anonymous feedback form, as well as a lot of discussion about future ways that we could organize and implement textbook groups.

Enerata has some corrections and some other things. Lastly, the active inference table section has a few tables that we reference often through the at key in coda that includes the active inference ontology.

Some links to other relevant research paper databases that we've compiled for other knowledge engineering work.

Some links to other platforms, as well as the live streams table, the several hundred live streams that we often reference as they discuss relevant papers and topics.

So, to kind of overview now, there's various ways to participate in cohort five, as well as cohort four.

And that can include your synchronous and or asynchronous participation. If you ever have any questions, you can reach out by email or you can tag by selecting some text and tagging active inference institute.

We will really look forward to your participation and good luck with the textbook group. Bye.